Radar range and velocity

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Separate a radar's range measurement from its line-of-sight velocity measurement.

SUCCESS CRITERIA - I can...

1. Use round-trip time for range.
2. Explain the information in a Doppler shift.
3. State what radial velocity leaves unknown.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "round trip" and "signal" to explain your model or result. Sentence starters are optional.

round trip: travel from start to a reflector and back

signal: a disturbance that carries information

Doppler shift: a frequency change associated with relative motion

01 Bring back an earlier idea

Two EM waves in vacuum have frequencies f and 3 f. **Compare** speeds and wavelengths.

Word help: vacuum = space with negligible matter

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: range = $c\Delta t/2$; $c = 3.0\times10^{8}$ m/s.
Radar time is a round trip. Doppler gives line-of-sight velocity information.

Worked reasoning: A radar echo returns after 2 microseconds. With $c=3\times10^{8}$ m/s, range$=c\Delta t/2=300$ m. The measured time includes the outward and return paths.

Your turn: Draw the round trip and label the conversion 2 microseconds$=2\times10^{-6}$ s.

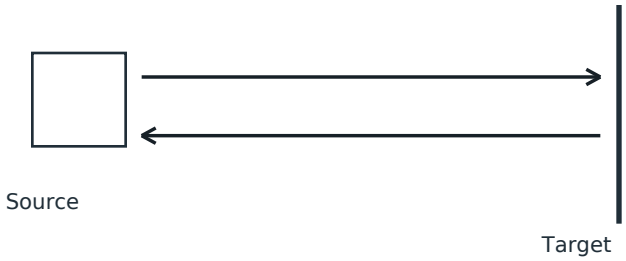
Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Look and explain

Draw outgoing and returning radar paths. Which measurement helps range: echo delay or frequency shift? Which helps radial velocity?

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: ① ② ③

04 Build the solution

In a toy model, signal speed 20 m/s and echo delay 3 s. **Find** total path and range.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

An asteroid has zero measured radial velocity at one instant. Does that prove it is stationary?
Sketch another possible motion.

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: 1 2 3

06 Choose a method

A wave has frequency 5 Hz and wavelength 7 m.
Find its speed.

Word help: wavelength = distance between corresponding points on repeated waves

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

07 Exit check

An echo’s out-and-back distance represents 1000 m. How far is the target?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Echo times change from 4 to 3 microseconds.
Calculate both ranges and state whether the target is approaching or receding.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3